

Lecture 6 Chain Rule

Tentative midterm schedule: Oct. 19th, Sunday, 7pm - 9:30pm.

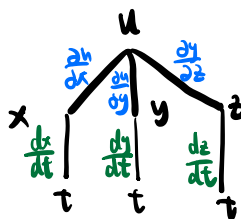
$$f(x), x(t) \rightarrow f(x(t)) \quad \frac{df}{dt} = \frac{df}{dx} \frac{dx}{dt}$$


$u(x, y, z)$ represents the temperature at the point (x, y, z)

A particle moves along the path $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$.

$u(\vec{r}(t))$ represents the temperature felt by the particle at time t

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt}$$



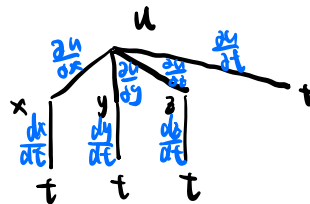
Example. Let $u(x, y, z) = x^2 + y^2 - 2z^2$ and $\langle x, y, z \rangle = \langle \cos t, \sin t, t \rangle$. Compute $\frac{du}{dt}$

Sol. $u = \cos^2 t + \sin^2 t - 2t^2 = 1 - 2t^2 \quad \frac{du}{dt} = -4t$

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt} \\ &= 2x(-\sin t) + 2y \cos t + (-4z) \cdot 1 \\ &= -2 \cos t \sin t + 2 \sin t \cos t - 4t \\ &= -4t \end{aligned}$$

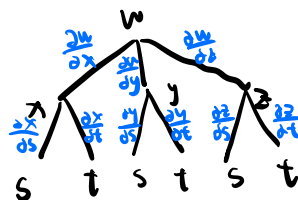
Temperature function $u(x, y, z, t)$, $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$

$$\boxed{\frac{du}{dt}} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt} + \boxed{\frac{\partial u}{\partial t}}$$



$$w(x, y, z), \quad x(s, t), \quad y(s, t), \quad z(s, t)$$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial s}$$



Example. $w = x + 2y + z^2$, $x = \frac{t}{s}$, $y = t^2 + \ln s$, $z = 2t$. Find $\frac{\partial w}{\partial s}$

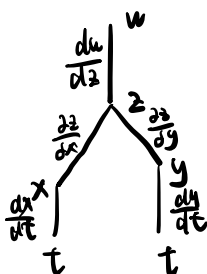
$$\text{Sol. } \frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial s}$$

$$= 1 \cdot \left(-\frac{t}{s^2}\right) + 2 \cdot \frac{1}{s} + 2z \cdot 0$$

$$= -\frac{t}{s^2} + \frac{2}{s}$$

$$w(z), \quad z(x, y), \quad x(t), \quad y(t)$$

$$\frac{dw}{dt} = \frac{dw}{dz} \frac{dz}{dx} \frac{dx}{dt} + \frac{dw}{dz} \frac{dz}{dy} \frac{dy}{dt}$$



Implicit Differentiation.

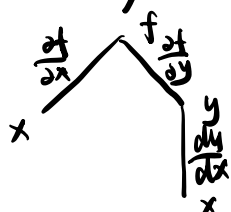
$$x^2 + y^3 + \sin^2 y = 1 \quad \frac{dy}{dx}$$

$$d(x^2 + y^3 + \sin^2 y) = 0$$

$$2x dx + 3y^2 dy + 2\sin y \cos y dy = 0$$

$$\frac{dy}{dx} = -\frac{2x}{3y^2 + 2\sin y \cos y}$$

Alternatively, let $f(x, y) = x^2 + y^3 + \sin^2 y = 1$



$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx}$$

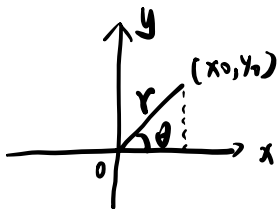
$$0 = f_x + f_y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{f_x}{f_y}$$

In this example, $f_x = 2x$, $f_y = 3y^2 + 2\sin y \cos y$

$$\Rightarrow \frac{dy}{dx} = -\frac{2x}{3y^2 + 2\sin y \cos y}$$

Chain Rule on Second Derivatives.

Polar Coordinates



$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \text{or} \quad \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \tan^{-1}(y/x) \end{cases}$$

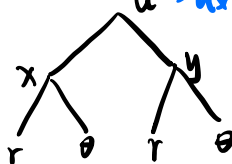
$$u(x, y) = u(r \cos \theta, r \sin \theta)$$

$$u = (\sqrt{x^2 + y^2})^3 \Rightarrow u = r^3$$

$$u(x, y), u_x, u_y, u_{xx}, u_{yy}, u_{xy} \Rightarrow \frac{\partial u}{\partial r}, \frac{\partial u}{\partial \theta}, \frac{\partial^2 u}{\partial r^2}, \frac{\partial^2 u}{\partial \theta^2}, \frac{\partial^2 u}{\partial \theta \partial r}$$

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial r}$$

$$= u_x \cos \theta + u_y \sin \theta$$



$$\frac{\partial^2 u}{\partial \theta \partial r} = \frac{\partial}{\partial \theta} (u_x \cos \theta + u_y \sin \theta)$$

$$= \frac{\partial u_x}{\partial \theta} \cos \theta - u_x \sin \theta + \frac{\partial u_y}{\partial \theta} \sin \theta + u_y \cos \theta.$$

$$\frac{\partial u_x}{\partial \theta} = \frac{\partial u_x}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial u_x}{\partial y} \frac{\partial y}{\partial \theta} = u_{xx} (-r \sin \theta) + u_{xy} (r \cos \theta)$$

$$\frac{\partial u_y}{\partial \theta} = \frac{\partial u_y}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial u_y}{\partial y} \frac{\partial y}{\partial \theta} = u_{xy} (-r \sin \theta) + u_{yy} (r \cos \theta)$$

$$\frac{\partial^2 u}{\partial \theta \partial r} = (-u_{xx} r \sin \theta + u_{xy} r \cos \theta) \cos \theta - u_x \sin \theta$$

$$+ (-u_{xy} r \sin \theta + u_{yy} r \cos \theta) \sin \theta + u_y \cos \theta.$$

$$= -u_{xx} r \sin \theta \cos \theta + u_{xy} r (\cos^2 \theta - \sin^2 \theta) + u_{yy} r \sin \theta \cos \theta - u_x \sin \theta + u_y \cos \theta.$$

Implicit Differentiation on Second-order Derivatives.

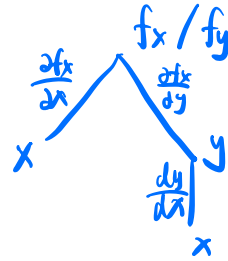
$$x^2 + y^3 + \sin^2 y = 1 \quad \frac{dy}{dx}, \quad \frac{d^2 y}{dx^2}$$

$$\frac{dy}{dx} = - \frac{2x}{3y^2 + 2 \sin y \cos y}$$

$f(x,y) = 0 \Rightarrow \frac{dy}{dx}$ as function of $f_x, f_y, f_{xx}, f_{xy}, f_{yy}$

Sol. $\frac{dy}{dx} = -\frac{f_x}{f_y}$

$$\frac{dy}{dx^2} = \frac{d}{dx} \left(-\frac{f_x}{f_y} \right) = - \frac{\frac{df_x}{dx} f_y - \frac{df}{dx} f_x}{f_y^2}$$



$$\frac{df_x}{dx} = \frac{\partial f_x}{\partial x} + \frac{\partial f_x}{\partial y} \frac{dy}{dx} = f_{xx} - f_{xy} \frac{f_x}{f_y}$$

$$\frac{df_y}{dx} = \frac{\partial f_y}{\partial x} + \frac{\partial f_y}{\partial y} \frac{dy}{dx} = f_{xy} - f_{yy} \frac{f_x}{f_y}$$

$$\frac{d^2y}{dx^2} = - \frac{(f_{xx} - f_{xy} \frac{f_x}{f_y}) f_y - (f_{xy} - f_{yy} \frac{f_x}{f_y}) f_x}{f_y^2}$$

$$= - \frac{f_{xx} f_y - 2 f_{xy} f_x + f_{yy} f_x^2 / f_y}{f_y^2}$$

$$= - \frac{f_{xx} f_y^2 - 2 f_{xy} f_x f_y + f_{yy} f_x^2}{f_y^3}$$

$$f(x,y) = x^2 + y^3 + \sin^2 y \Rightarrow f_x = 2x, \quad f_y = 3y^2 + 2\sin y \cos y = 3y^2 + \sin 2y$$

$$f_{xx} = 2, \quad f_{xy} = 0, \quad f_{yy} = 6y + 2\cos 2y$$

$$\frac{d^2y}{dx^2} = - \frac{2(3y^2 + \sin 2y)^2 + (6y + 2\cos 2y)x^2}{(3y^2 + \sin 2y)^3}$$